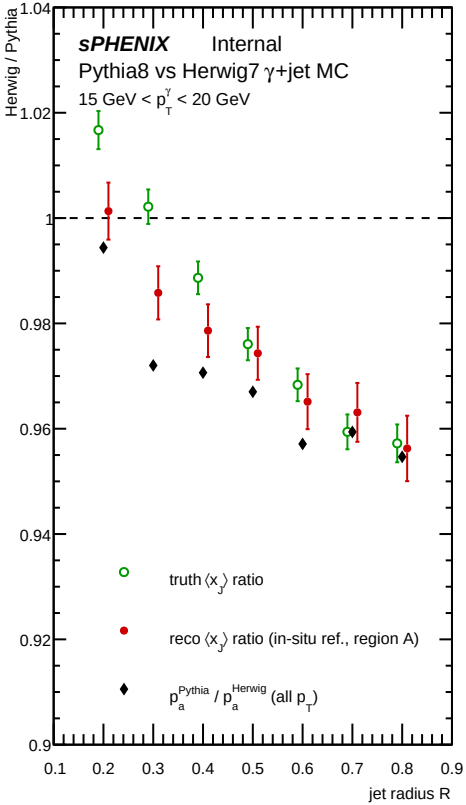
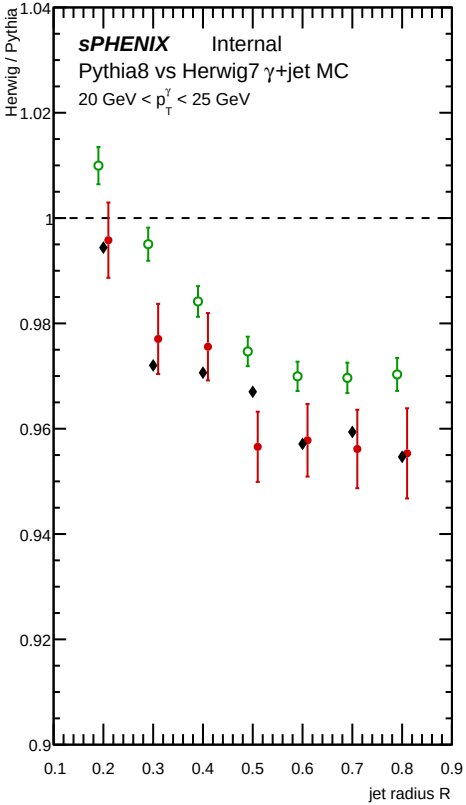
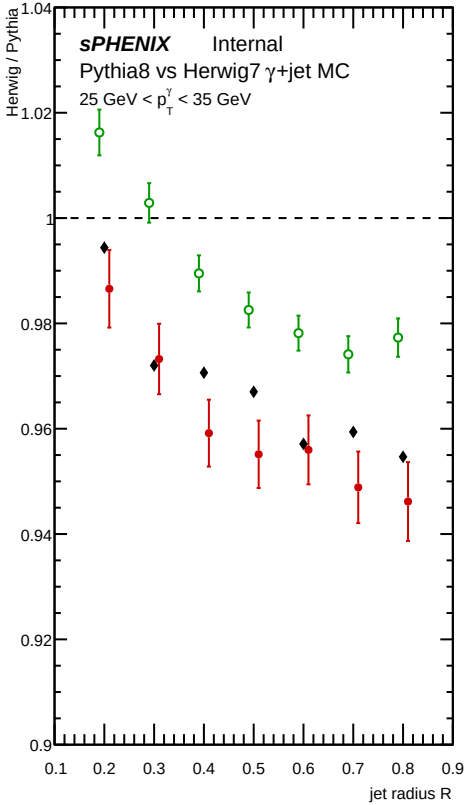
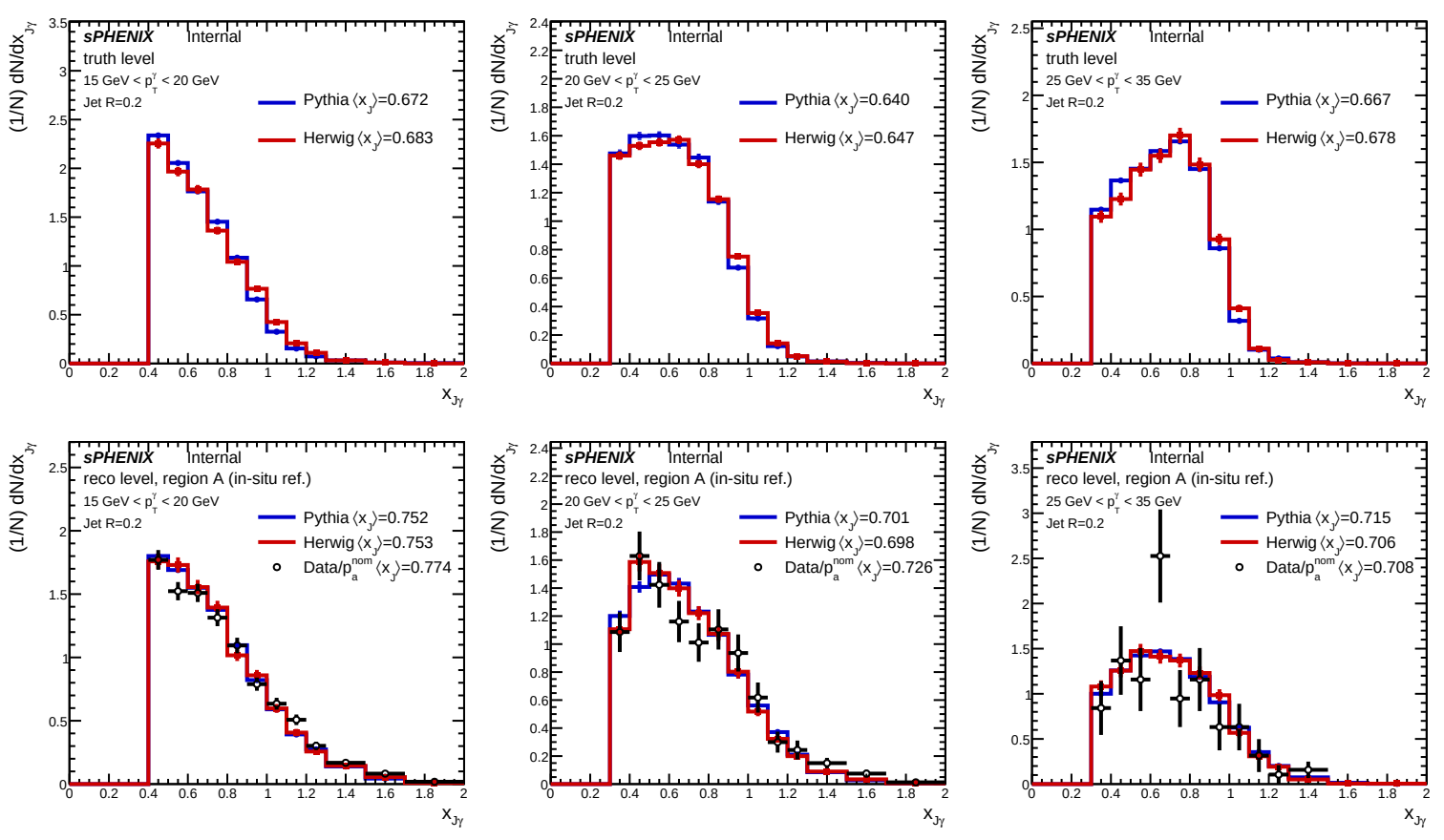
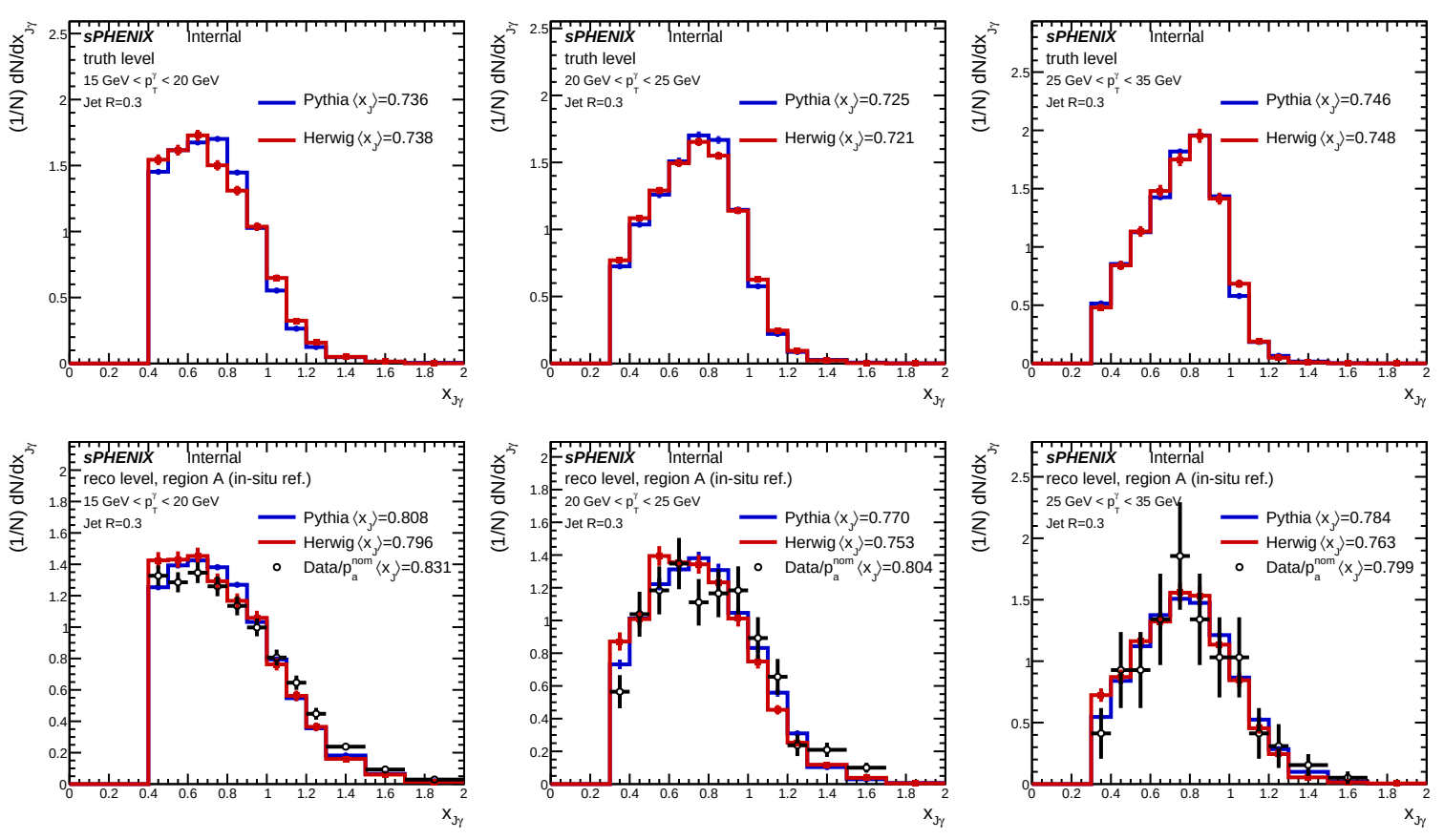
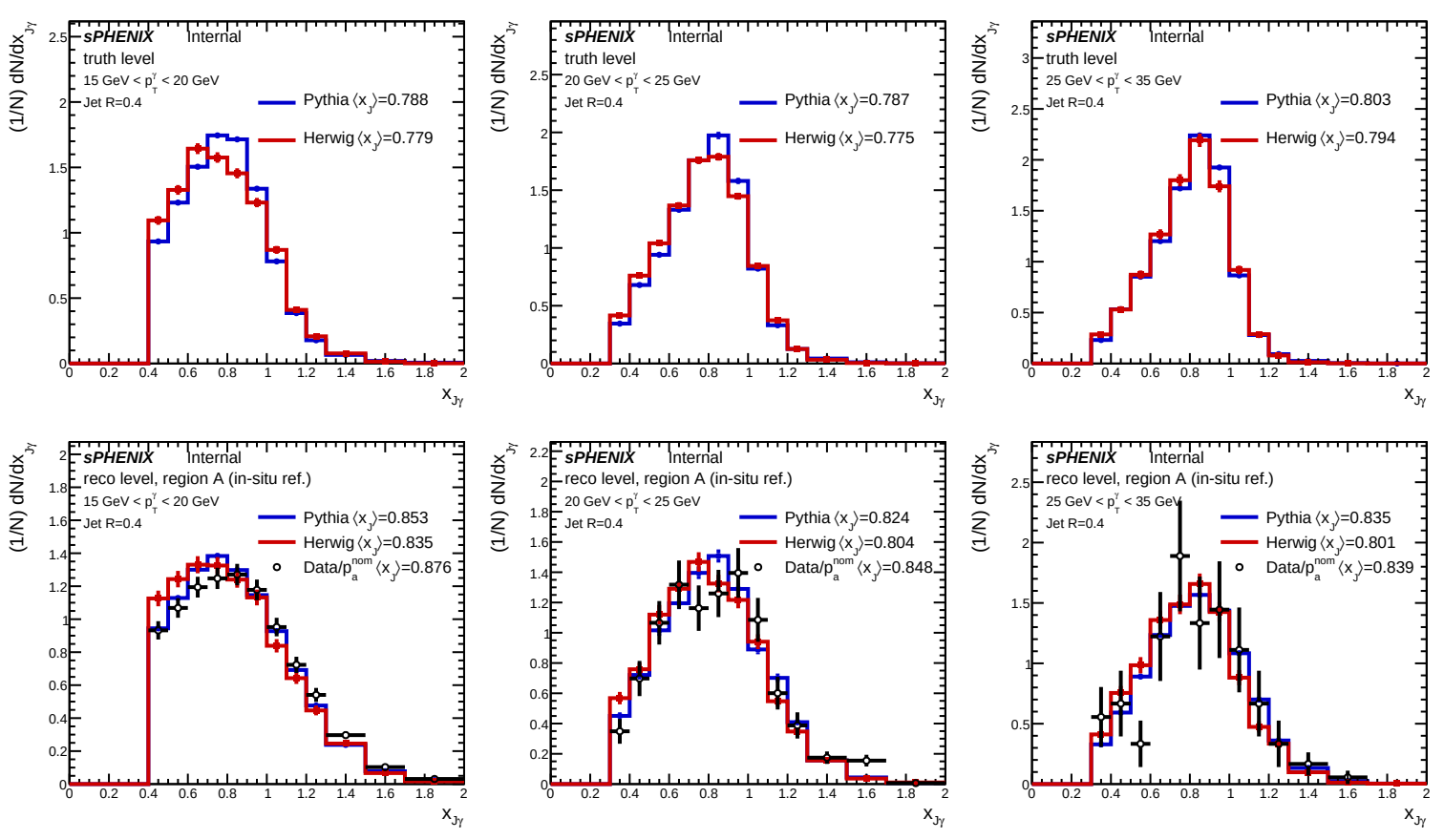

sPHENIX Internal
Pythia8 vs Herwig7 γ+jet MC
15 GeV < p_T^γ < 20 GeV
truth ⟨x_J⟩ ratio
reco ⟨x_J⟩ ratio (in-situ ref., region A)
p_a^Pythia / p_a^Herwig (all p_T)
Herwig / Pythia
jet radius R

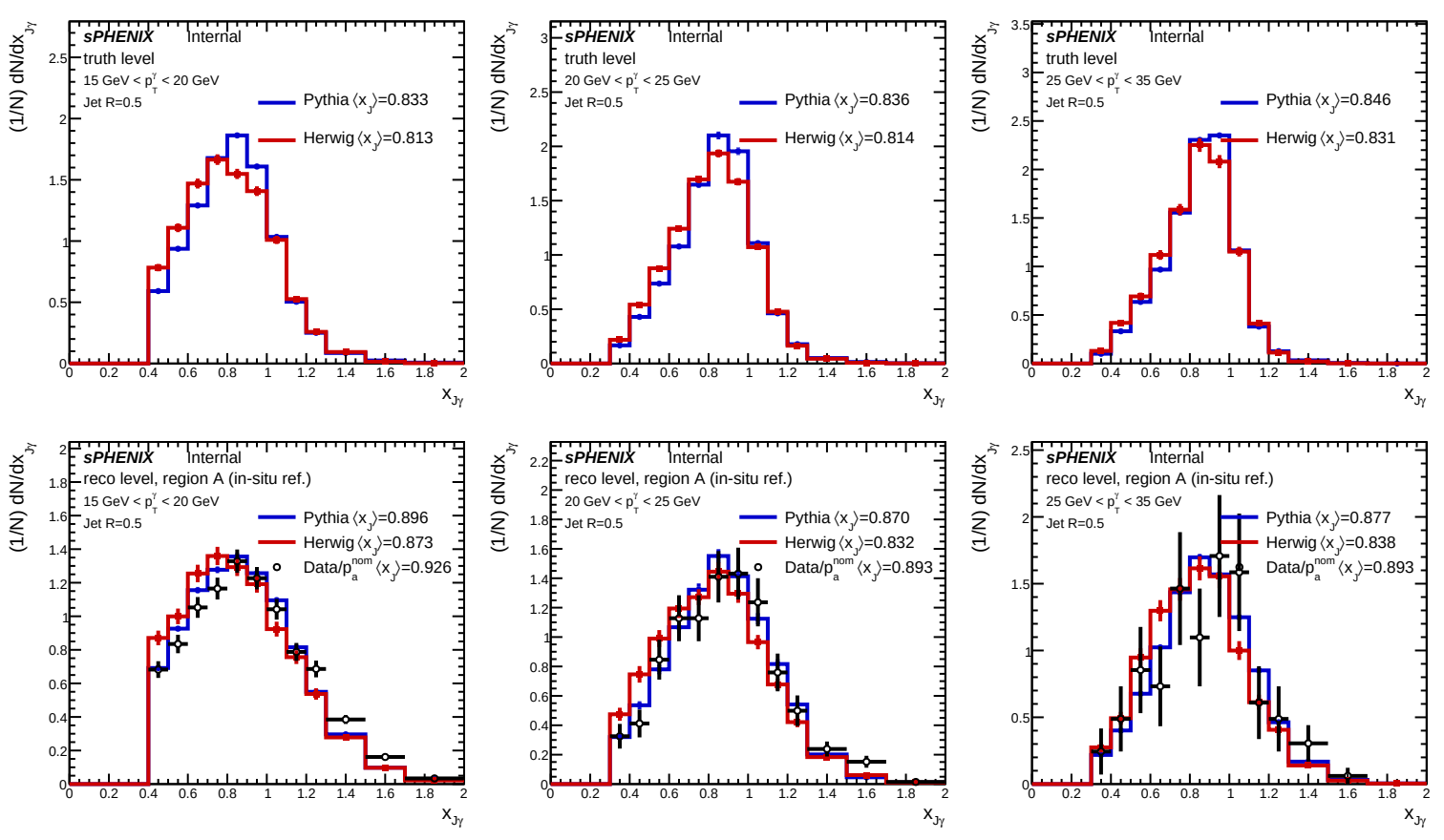

sPHENIX Internal
Pythia8 vs Herwig7 γ+jet MC
20 GeV < p_T^γ < 25 GeV
Herwig / Pythia
jet radius R

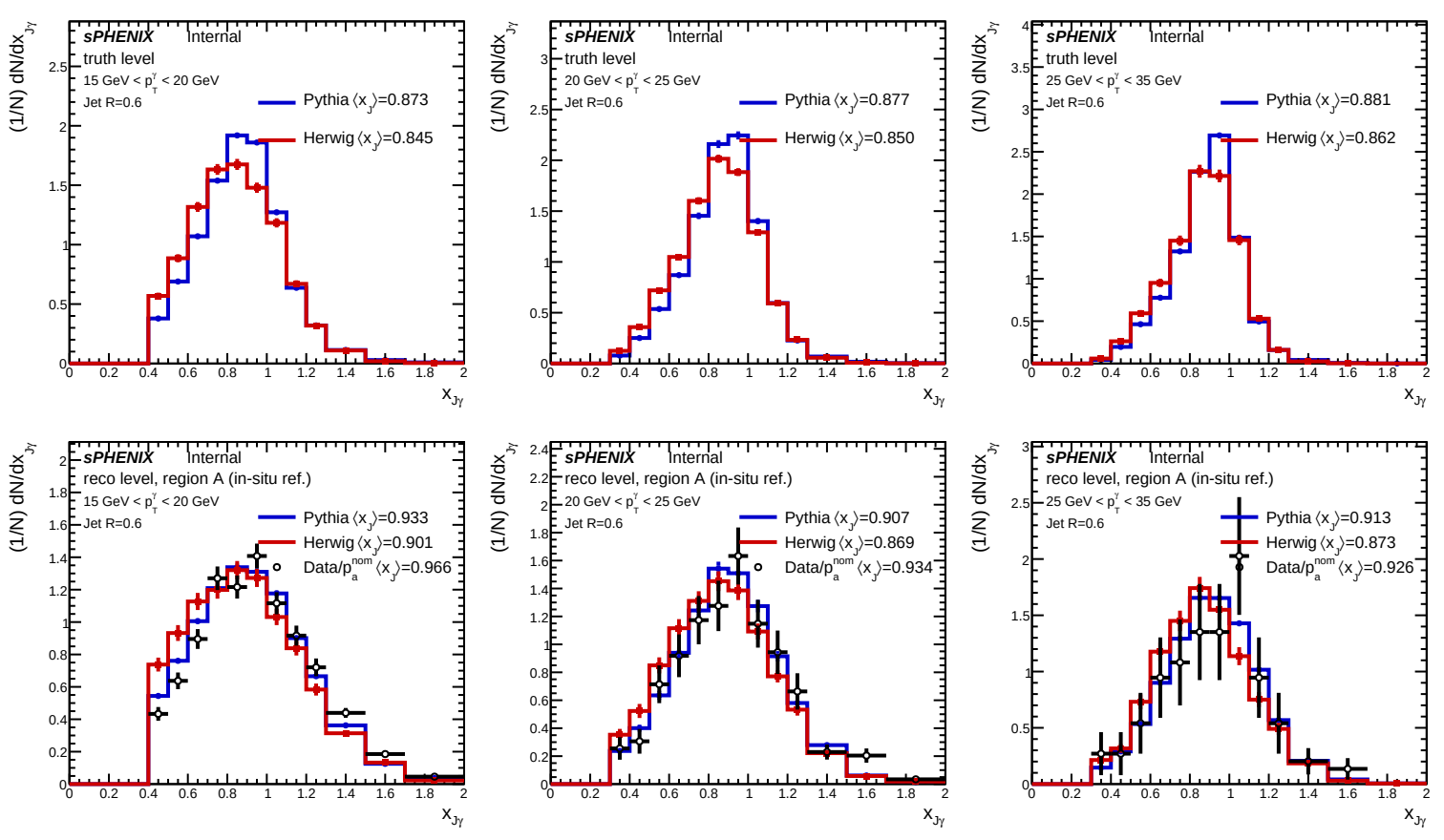

sPHENIX Internal
Pythia8 vs Herwig7 γ+jet MC
25 GeV < p_T^γ < 35 GeV
Herwig / Pythia
jet radius R

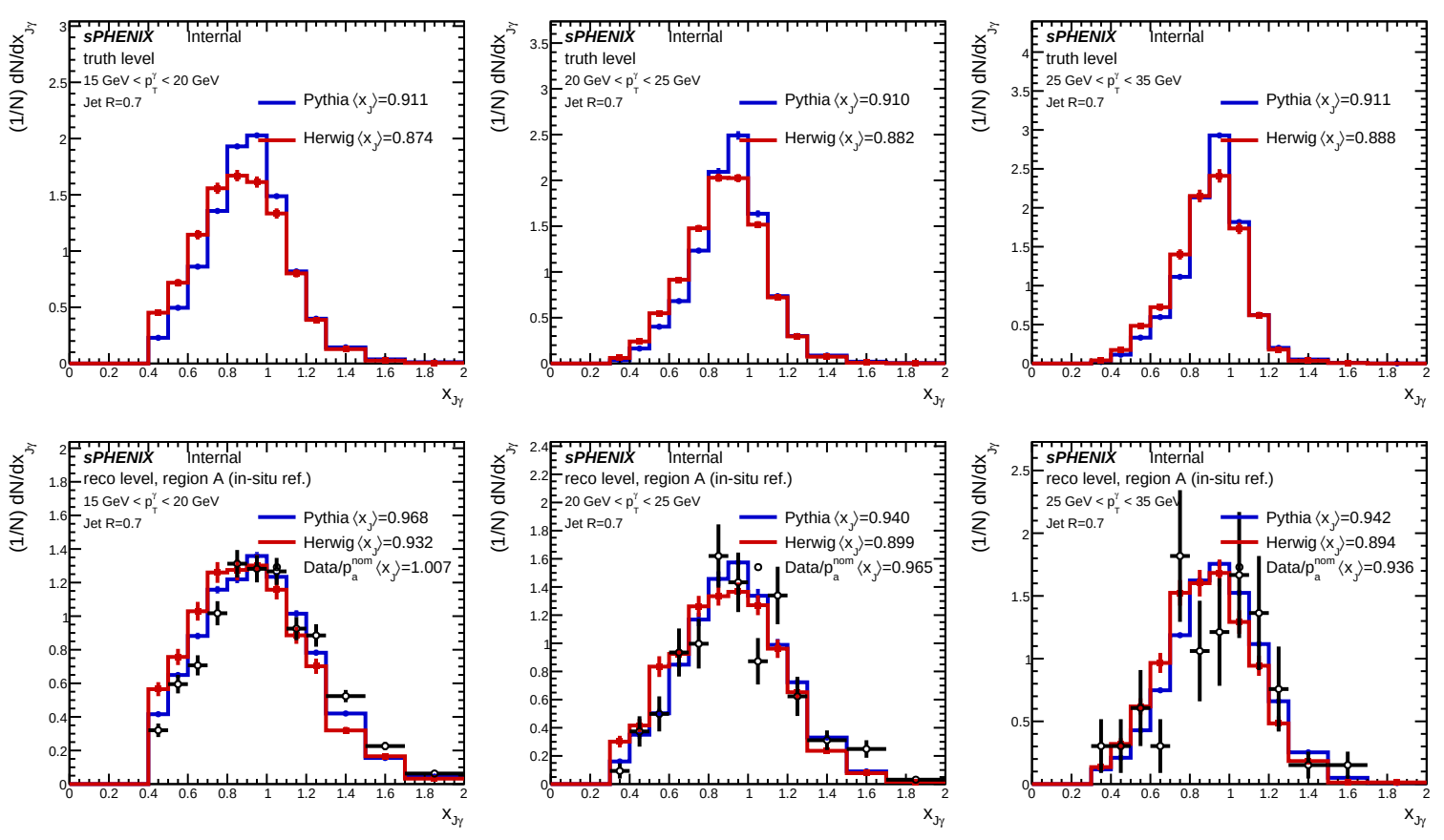

sPHENIX Internal
truth level
15 GeV < $p_T^\gamma$ < 20 GeV
Jet R=0.2
Pythia $\langle x_{J\gamma}\rangle$=0.672
Herwig $\langle x_{J\gamma}\rangle$=0.683
sPHENIX Internal
truth level
20 GeV < $p_T^\gamma$ < 25 GeV
Jet R=0.2
Pythia $\langle x_{J\gamma}\rangle$=0.640
Herwig $\langle x_{J\gamma}\rangle$=0.647
sPHENIX Internal
truth level
25 GeV < $p_T^\gamma$ < 35 GeV
Jet R=0.2
Pythia $\langle x_{J\gamma}\rangle$=0.667
Herwig $\langle x_{J\gamma}\rangle$=0.678
sPHENIX Internal
reco level, region A (in-situ ref.)
15 GeV < $p_T^\gamma$ < 20 GeV
Jet R=0.2
Pythia $\langle x_{J\gamma}\rangle$=0.752
Herwig $\langle x_{J\gamma}\rangle$=0.753
Data/$p_a^{nom}$ $\langle x_{J\gamma}\rangle$=0.774
sPHENIX Internal
reco level, region A (in-situ ref.)
20 GeV < $p_T^\gamma$ < 25 GeV
Jet R=0.2
Pythia $\langle x_{J\gamma}\rangle$=0.701
Herwig $\langle x_{J\gamma}\rangle$=0.698
Data/$p_a^{nom}$ $\langle x_{J\gamma}\rangle$=0.726
sPHENIX Internal
reco level, region A (in-situ ref.)
25 GeV < $p_T^\gamma$ < 35 GeV
Jet R=0.2
Pythia $\langle x_{J\gamma}\rangle$=0.715
Herwig $\langle x_{J\gamma}\rangle$=0.706
Data/$p_a^{nom}$ $\langle x_{J\gamma}\rangle$=0.708
(1/N) dN/dx$_{J\gamma}$
$x_{J\gamma}$

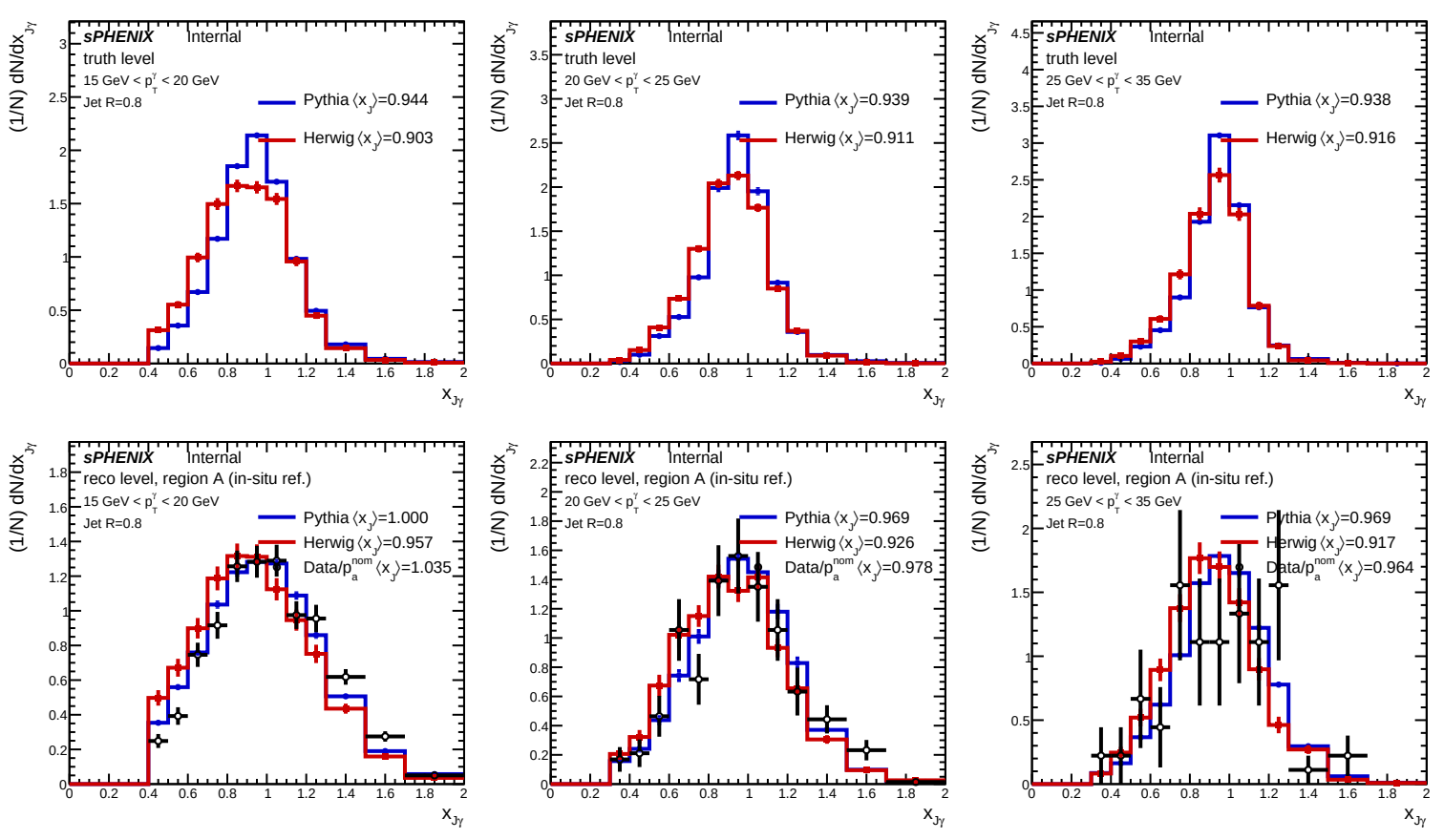

sPHENIX Internal
truth level
15 GeV < $p_T^\gamma$ < 20 GeV
Jet R=0.3
Pythia $\langle x_{J\gamma}\rangle$=0.736
Herwig $\langle x_{J\gamma}\rangle$=0.738
sPHENIX Internal
truth level
20 GeV < $p_T^\gamma$ < 25 GeV
Jet R=0.3
Pythia $\langle x_{J\gamma}\rangle$=0.725
Herwig $\langle x_{J\gamma}\rangle$=0.721
sPHENIX Internal
truth level
25 GeV < $p_T^\gamma$ < 35 GeV
Jet R=0.3
Pythia $\langle x_{J\gamma}\rangle$=0.746
Herwig $\langle x_{J\gamma}\rangle$=0.748
sPHENIX Internal
reco level, region A (in-situ ref.)
15 GeV < $p_T^\gamma$ < 20 GeV
Jet R=0.3
Pythia $\langle x_{J\gamma}\rangle$=0.808
Herwig $\langle x_{J\gamma}\rangle$=0.796
Data/$p_a^{nom}$ $\langle x_{J\gamma}\rangle$=0.831
sPHENIX Internal
reco level, region A (in-situ ref.)
20 GeV < $p_T^\gamma$ < 25 GeV
Jet R=0.3
Pythia $\langle x_{J\gamma}\rangle$=0.770
Herwig $\langle x_{J\gamma}\rangle$=0.753
Data/$p_a^{nom}$ $\langle x_{J\gamma}\rangle$=0.804
sPHENIX Internal
reco level, region A (in-situ ref.)
25 GeV < $p_T^\gamma$ < 35 GeV
Jet R=0.3
Pythia $\langle x_{J\gamma}\rangle$=0.784
Herwig $\langle x_{J\gamma}\rangle$=0.763
Data/$p_a^{nom}$ $\langle x_{J\gamma}\rangle$=0.799
(1/N) dN/dx$_{J\gamma}$
$x_{J\gamma}$

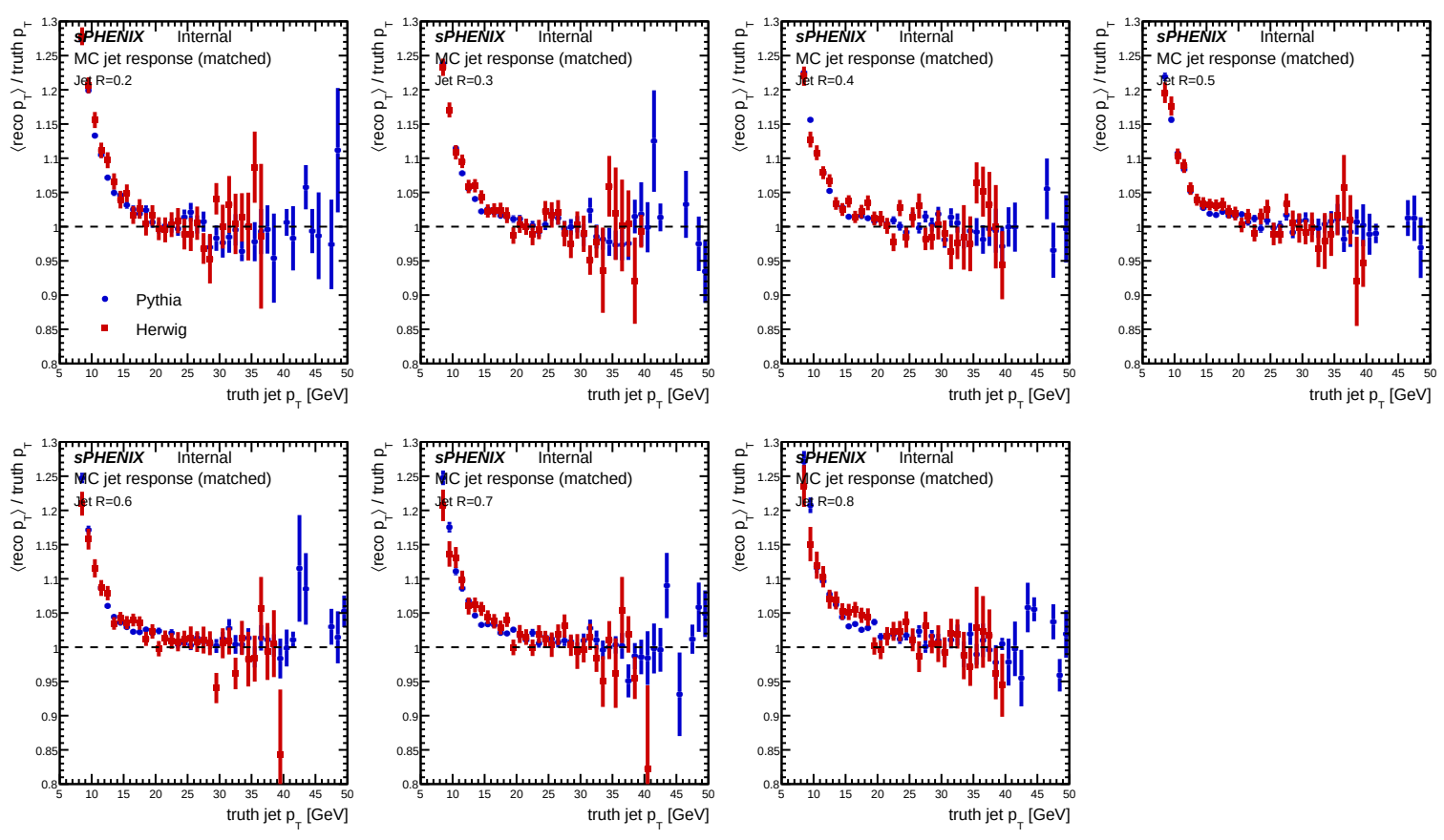

sPHENIX Internal
truth level
15 GeV < $p_T^\gamma$ < 20 GeV
Jet R=0.4
Pythia $\langle x_{J\gamma}\rangle$=0.788
Herwig $\langle x_{J\gamma}\rangle$=0.779
sPHENIX Internal
truth level
20 GeV < $p_T^\gamma$ < 25 GeV
Jet R=0.4
Pythia $\langle x_{J\gamma}\rangle$=0.787
Herwig $\langle x_{J\gamma}\rangle$=0.775
sPHENIX Internal
truth level
25 GeV < $p_T^\gamma$ < 35 GeV
Jet R=0.4
Pythia $\langle x_{J\gamma}\rangle$=0.803
Herwig $\langle x_{J\gamma}\rangle$=0.794
sPHENIX Internal
reco level, region A (in-situ ref.)
15 GeV < $p_T^\gamma$ < 20 GeV
Jet R=0.4
Pythia $\langle x_{J\gamma}\rangle$=0.853
Herwig $\langle x_{J\gamma}\rangle$=0.835
Data/$p_a^{nom}$ $\langle x_{J\gamma}\rangle$=0.876
sPHENIX Internal
reco level, region A (in-situ ref.)
20 GeV < $p_T^\gamma$ < 25 GeV
Jet R=0.4
Pythia $\langle x_{J\gamma}\rangle$=0.824
Herwig $\langle x_{J\gamma}\rangle$=0.804
Data/$p_a^{nom}$ $\langle x_{J\gamma}\rangle$=0.848
sPHENIX Internal
reco level, region A (in-situ ref.)
25 GeV < $p_T^\gamma$ < 35 GeV
Jet R=0.4
Pythia $\langle x_{J\gamma}\rangle$=0.835
Herwig $\langle x_{J\gamma}\rangle$=0.801
Data/$p_a^{nom}$ $\langle x_{J\gamma}\rangle$=0.839
(1/N) dN/dx$_{J\gamma}$
$x_{J\gamma}$

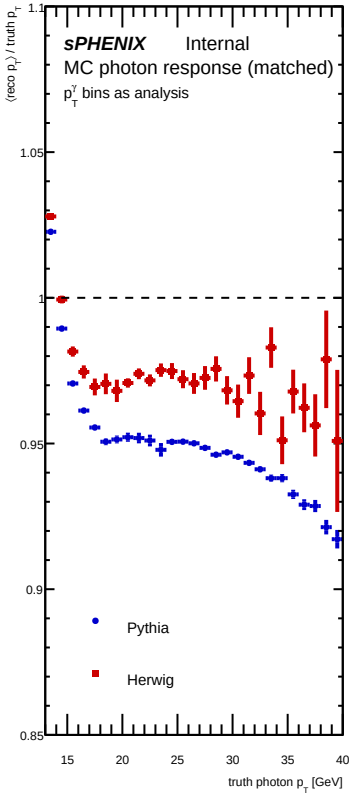

sPHENIX Internal
truth level
15 GeV < $p_T^\gamma$ < 20 GeV
Jet R=0.5
Pythia ⟨$x_{J\gamma}$⟩=0.833
Herwig ⟨$x_{J\gamma}$⟩=0.813
(1/N) dN/d$x_{J\gamma}$
$x_{J\gamma}$
sPHENIX Internal
truth level
20 GeV < $p_T^\gamma$ < 25 GeV
Jet R=0.5
Pythia ⟨$x_{J\gamma}$⟩=0.836
Herwig ⟨$x_{J\gamma}$⟩=0.814
(1/N) dN/d$x_{J\gamma}$
$x_{J\gamma}$
sPHENIX Internal
truth level
25 GeV < $p_T^\gamma$ < 35 GeV
Jet R=0.5
Pythia ⟨$x_{J\gamma}$⟩=0.846
Herwig ⟨$x_{J\gamma}$⟩=0.831
(1/N) dN/d$x_{J\gamma}$
$x_{J\gamma}$
sPHENIX Internal
reco level, region A (in-situ ref.)
15 GeV < $p_T^\gamma$ < 20 GeV
Jet R=0.5
Pythia ⟨$x_{J\gamma}$⟩=0.896
Herwig ⟨$x_{J\gamma}$⟩=0.873
Data/$p_a^{nom}$ ⟨$x_{J\gamma}$⟩=0.926
(1/N) dN/d$x_{J\gamma}$
$x_{J\gamma}$
sPHENIX Internal
reco level, region A (in-situ ref.)
20 GeV < $p_T^\gamma$ < 25 GeV
Jet R=0.5
Pythia ⟨$x_{J\gamma}$⟩=0.870
Herwig ⟨$x_{J\gamma}$⟩=0.832
Data/$p_a^{nom}$ ⟨$x_{J\gamma}$⟩=0.893
(1/N) dN/d$x_{J\gamma}$
$x_{J\gamma}$
sPHENIX Internal
reco level, region A (in-situ ref.)
25 GeV < $p_T^\gamma$ < 35 GeV
Jet R=0.5
Pythia ⟨$x_{J\gamma}$⟩=0.877
Herwig ⟨$x_{J\gamma}$⟩=0.838
Data/$p_a^{nom}$ ⟨$x_{J\gamma}$⟩=0.893
(1/N) dN/d$x_{J\gamma}$
$x_{J\gamma}$

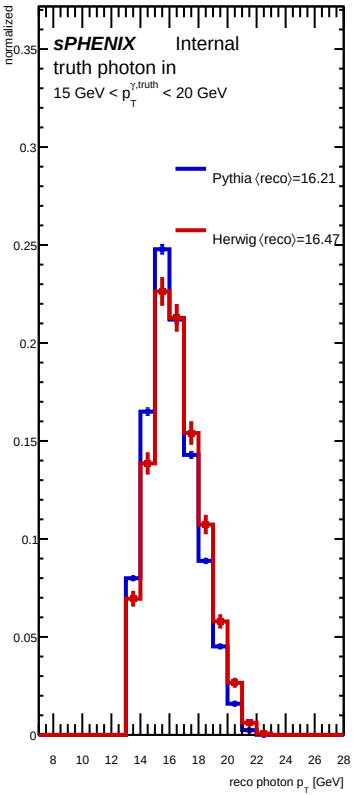

sPHENIX Internal
truth level
15 GeV < $p_T^\gamma$ < 20 GeV
Jet R=0.6
Pythia $\langle x_{J\gamma}\rangle$=0.873
Herwig $\langle x_{J\gamma}\rangle$=0.845
sPHENIX Internal
truth level
20 GeV < $p_T^\gamma$ < 25 GeV
Jet R=0.6
Pythia $\langle x_{J\gamma}\rangle$=0.877
Herwig $\langle x_{J\gamma}\rangle$=0.850
sPHENIX Internal
truth level
25 GeV < $p_T^\gamma$ < 35 GeV
Jet R=0.6
Pythia $\langle x_{J\gamma}\rangle$=0.881
Herwig $\langle x_{J\gamma}\rangle$=0.862
sPHENIX Internal
reco level, region A (in-situ ref.)
15 GeV < $p_T^\gamma$ < 20 GeV
Jet R=0.6
Pythia $\langle x_{J\gamma}\rangle$=0.933
Herwig $\langle x_{J\gamma}\rangle$=0.901
Data/$p_a^{nom}$ $\langle x_{J\gamma}\rangle$=0.966
sPHENIX Internal
reco level, region A (in-situ ref.)
20 GeV < $p_T^\gamma$ < 25 GeV
Jet R=0.6
Pythia $\langle x_{J\gamma}\rangle$=0.907
Herwig $\langle x_{J\gamma}\rangle$=0.869
Data/$p_a^{nom}$ $\langle x_{J\gamma}\rangle$=0.934
sPHENIX Internal
reco level, region A (in-situ ref.)
25 GeV < $p_T^\gamma$ < 35 GeV
Jet R=0.6
Pythia $\langle x_{J\gamma}\rangle$=0.913
Herwig $\langle x_{J\gamma}\rangle$=0.873
Data/$p_a^{nom}$ $\langle x_{J\gamma}\rangle$=0.926
(1/N) dN/dx$_{J\gamma}$
$x_{J\gamma}$

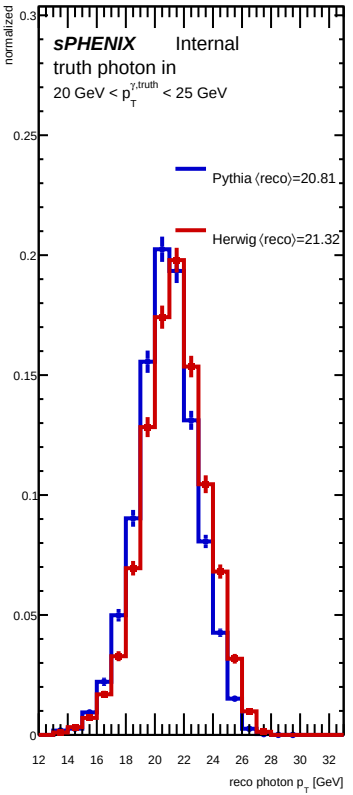

sPHENIX Internal
truth level
15 GeV < p_T^γ < 20 GeV
Jet R=0.7
Pythia ⟨x_Jγ⟩=0.911
Herwig ⟨x_Jγ⟩=0.874
(1/N) dN/dx_Jγ
x_Jγ
sPHENIX Internal
truth level
20 GeV < p_T^γ < 25 GeV
Jet R=0.7
Pythia ⟨x_Jγ⟩=0.910
Herwig ⟨x_Jγ⟩=0.882
(1/N) dN/dx_Jγ
x_Jγ
sPHENIX Internal
truth level
25 GeV < p_T^γ < 35 GeV
Jet R=0.7
Pythia ⟨x_Jγ⟩=0.911
Herwig ⟨x_Jγ⟩=0.888
(1/N) dN/dx_Jγ
x_Jγ
sPHENIX Internal
reco level, region A (in-situ ref.)
15 GeV < p_T^γ < 20 GeV
Jet R=0.7
Pythia ⟨x_Jγ⟩=0.968
Herwig ⟨x_Jγ⟩=0.932
Data/p_a^nom ⟨x_Jγ⟩=1.007
(1/N) dN/dx_Jγ
x_Jγ
sPHENIX Internal
reco level, region A (in-situ ref.)
20 GeV < p_T^γ < 25 GeV
Jet R=0.7
Pythia ⟨x_Jγ⟩=0.940
Herwig ⟨x_Jγ⟩=0.899
Data/p_a^nom ⟨x_Jγ⟩=0.965
(1/N) dN/dx_Jγ
x_Jγ
sPHENIX Internal
reco level, region A (in-situ ref.)
25 GeV < p_T^γ < 35 GeV
Jet R=0.7
Pythia ⟨x_Jγ⟩=0.942
Herwig ⟨x_Jγ⟩=0.894
Data/p_a^nom ⟨x_Jγ⟩=0.936
(1/N) dN/dx_Jγ
x_Jγ

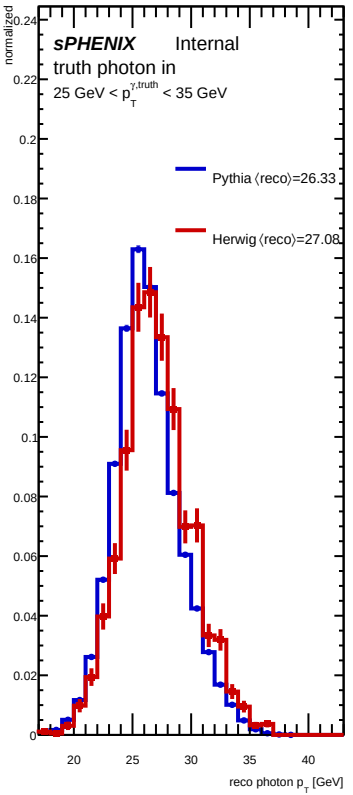

sPHENIX Internal
truth level
15 GeV < $p_T^\gamma$ < 20 GeV
Jet R=0.8
Pythia ⟨$x_{J\gamma}$⟩=0.944
Herwig ⟨$x_{J\gamma}$⟩=0.903
(1/N) dN/d$x_{J\gamma}$
$x_{J\gamma}$
sPHENIX Internal
truth level
20 GeV < $p_T^\gamma$ < 25 GeV
Jet R=0.8
Pythia ⟨$x_{J\gamma}$⟩=0.939
Herwig ⟨$x_{J\gamma}$⟩=0.911
(1/N) dN/d$x_{J\gamma}$
$x_{J\gamma}$
sPHENIX Internal
truth level
25 GeV < $p_T^\gamma$ < 35 GeV
Jet R=0.8
Pythia ⟨$x_{J\gamma}$⟩=0.938
Herwig ⟨$x_{J\gamma}$⟩=0.916
(1/N) dN/d$x_{J\gamma}$
$x_{J\gamma}$
sPHENIX Internal
reco level, region A (in-situ ref.)
15 GeV < $p_T^\gamma$ < 20 GeV
Jet R=0.8
Pythia ⟨$x_{J\gamma}$⟩=1.000
Herwig ⟨$x_{J\gamma}$⟩=0.957
Data/$p_a^{nom}$ ⟨$x_{J\gamma}$⟩=1.035
(1/N) dN/d$x_{J\gamma}$
$x_{J\gamma}$
sPHENIX Internal
reco level, region A (in-situ ref.)
20 GeV < $p_T^\gamma$ < 25 GeV
Jet R=0.8
Pythia ⟨$x_{J\gamma}$⟩=0.969
Herwig ⟨$x_{J\gamma}$⟩=0.926
Data/$p_a^{nom}$ ⟨$x_{J\gamma}$⟩=0.978
(1/N) dN/d$x_{J\gamma}$
$x_{J\gamma}$
sPHENIX Internal
reco level, region A (in-situ ref.)
25 GeV < $p_T^\gamma$ < 35 GeV
Jet R=0.8
Pythia ⟨$x_{J\gamma}$⟩=0.969
Herwig ⟨$x_{J\gamma}$⟩=0.917
Data/$p_a^{nom}$ ⟨$x_{J\gamma}$⟩=0.964
(1/N) dN/d$x_{J\gamma}$
$x_{J\gamma}$


sPHENIX Internal
MC jet response (matched)
Jet R=0.2
Pythia
Herwig
⟨reco $p_T$⟩ / truth $p_T$
truth jet $p_T$ [GeV]
sPHENIX Internal
MC jet response (matched)
Jet R=0.3
sPHENIX Internal
MC jet response (matched)
Jet R=0.4
sPHENIX Internal
MC jet response (matched)
Jet R=0.5
sPHENIX Internal
MC jet response (matched)
Jet R=0.6
sPHENIX Internal
MC jet response (matched)
Jet R=0.7
sPHENIX Internal
MC jet response (matched)
Jet R=0.8


sPHENIX Internal
MC photon response (matched)
$p_T^\gamma$ bins as analysis
Pythia
Herwig
$\langle$reco $p_T\rangle$ / truth $p_T$
truth photon $p_T$ [GeV]


sPHENIX Internal
truth photon in
15 GeV < $p_T^{\gamma,truth}$ < 20 GeV
Pythia $\langle$reco$\rangle$=16.21
Herwig $\langle$reco$\rangle$=16.47
normalized
reco photon $p_T$ [GeV]


sPHENIX Internal
truth photon in
20 GeV < $p_T^{\gamma,truth}$ < 25 GeV
Pythia $\langle$reco$\rangle$=20.81
Herwig $\langle$reco$\rangle$=21.32
normalized
reco photon $p_T$ [GeV]


sPHENIX Internal
truth photon in
25 GeV < $p_T^{\gamma,truth}$ < 35 GeV
Pythia $\langle$reco$\rangle$=26.33
Herwig $\langle$reco$\rangle$=27.08
normalized
reco photon $p_T$ [GeV]